

Study

Anki flashcards on the Xteink X4 Pro. FSRS scheduling, your own decks, your own fonts. Reviews sync back to Anki.

The daily loop

1. Put a deck on the card

Convert from your live Anki collection, copy to the SD card.

2. Study

Apps > STUDY. Tap to reveal, tap a button to grade.

3. Sync back

One command applies your reviews and pushes to AnkiWeb.

The two commands

Run from `firmware-next/`. Both work out what they can and ask once for what they cannot, then remember it.

Once -- puts a deck and its fonts on the card

```
./tools_local/study/study.py setup
```

Finds your Anki profile, lists your decks so you pick one, finds the SD card, converts, and builds the fonts only if they are missing or no longer cover the deck. Run it again to switch deck.

After every session -- puts your reviews back into Anki

```
./tools_local/study/study.py sync --ankiweb
```

Refuses to run while Anki is open, backs the collection up first, and is safe to re-run. Drop `--ankiweb` to apply locally only; add `--dry-run` to look before you leap, or `--reconvert` to refresh the card afterwards.

Any time -- what is set up, and what is waiting

```
./tools_local/study/study.py status
```

Underneath are four single-purpose tools (`anki_to_deck`, `deck_to_anki`, `make_fonts`, `check_deck`) that take explicit paths. Reach for them when something goes wrong or you want one step only.

On the SD card

`/study/mandarin/` the deck (~70 KB)

`/study/fonts/` five CJK faces (~32 MB)

deck.dat card text, never written by the device.

cards.dat scheduling state.

revlog.dat your reviews, waiting to sync.

Retraining FSRS

There is no optimiser on the device and there should not be. Sync your reviews, then in Anki run **Deck options > FSRS > Optimize**. Then run `study.py setup` again -- the new weights travel with the deck.

Good to know

Fonts change every card. Five faces, picked at random, so you never lock onto one letterform. Same five your Anki template uses.

Scheduling matches Anki. Same FSRS weights, same learning steps, read from your deck preset. Individual intervals differ by a few percent only because Anki randomises its own and this does not.

Reviews are never lost. Every answer is flushed to the card immediately, and sync is idempotent.

Back leaves at any point. Everything else is a tap.

Not there yet

Sentence images 290 of your 301 cards have one. The device shows text only, so those cards are missing a cue Anki gives you. The biggest gap.

Real hardware Everything so far is verified in the simulator through the real render path. Untested on the device itself.

One deck at a time The app reads `/study/mandarin`. Switching decks means reconverting.

Leech tags Your leech action is Tag Only, so Anki keeps showing them too -- the device already matches. Only the tag itself is not added here; lapses still sync.

No editing No adding, editing, undo, or custom study. Do those in Anki.

No audio The X4 Pro has no speaker. Permanent.

Full detail: [docs/study-deck-format.md](#) · scheduler notes in `src/apps_local/study/StudyFsrs.h`